Q1

You are designing a spam email classifier using binary features (e.g., has_link: yes/no, urgent_words: yes/no). There are 6 binary features, and identical feature vectors always map to the same class (i.e., if $\mathbf{x}^{(1)} = \mathbf{x}^{(2)}$, then $y^{(1)} = y^{(2)}$).

a. How many distinct data points suffice for a perfect classifier (accuracy=1)? (1M)
b. With exactly that many distinct points, compare building a Naive Bayes vs. CART decision tree—which is more efficient in training time, and why? (2M)
c. For k-NN classification on these points (the data points used in Q 1.a) , what k value guarantees perfect accuracy on the training set? Explain briefly. (2M)

Q2. (10M)

Your colleague tracks daily productivity (High, Low) over 15 days. Attributes: Temperature (Hot, Mild), Caffeine (High, Low), Meetings (Many, Few), Deadline (Yes, No). Build ID3 decision tree to predict Productivity.

Obs	Temp	Caffeine	Meetings	Deadline	Productivity
1	Hot	High	Many	No	Low
2	Hot	High	Many	Yes	High
3	Hot	Low	Many	Yes	Low
4	Hot	Low	Many	No	Low
5	Hot	High	Few	Yes	High
6	Hot	High	Few	No	High
7	Hot	Low	Few	Yes	Low
8	Hot	Low	Few	No	Low
9	Mild	High	Many	Yes	High
10	Mild	High	Many	No	High
11	Mild	Low	Many	Yes	Low
12	Mild	Low	Many	No	Low
13	Mild	High	Few	Yes	High
14	Mild	High	Few	No	High
15	Mild	Low	Few	Yes	High

Show steps: entropy calculations, splits, and final tree.

Q3

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Answer only True (T) or False (F). No justification. Correct: +1M, Incorrect: -1M.

a. In boosting (e.g., AdaBoost) for binary classification, weak learners with accuracy <0.5 cannot be used to boost overall performance. F

b. K-means guarantees the global optimal clustering regardless of initialization. F

c. Apriori prunes candidate itemsets using only the anti-monotone property. T

d. ID3 can handle continuous attributes without discretization. F

e. Naive Bayes assumes feature independence given the class, making it equivalent to full Bayes when true. T

Rough Work